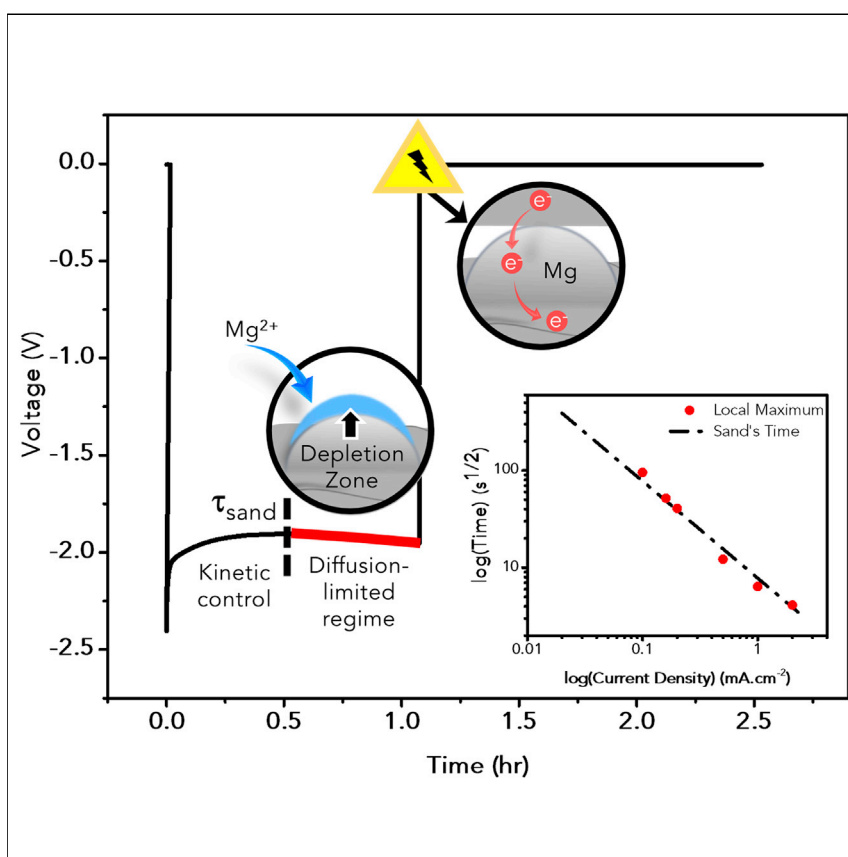


Article

Kinetic- versus Diffusion-Driven Three-Dimensional Growth in Magnesium Metal Battery Anodes



With twice the volumetric energy density as lithium, magnesium is a promising material for next-generation energy storage devices. Although magnesium rechargeable batteries were once believed to be even safer than lithium, the mechanisms governing multivalent metal deposition under practical operating conditions are still poorly understood. Through a comprehensive study of electrodeposition in coin cells over a wide range of current densities, we report a transition from charge-transfer-limited to diffusion-limited processes at timescales defined by the transport properties of the electrolyte. Our results explain controversial ideas such as 3D growth within the context of classical electrochemical theories and lay groundwork for future approaches to achieve stable electroplated multivalent metal electrodes.

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HIGHLIGHTS

3D Mg growth arises from mixed kinetic and diffusion control during electroplating

Electrodeposition terminates in short circuit at all currents studied (0.2–2 mA/cm²)

Traditional separators act as scaffolds for magnesium metal growth, not barriers

Article

Kinetic- versus Diffusion-Driven Three-Dimensional Growth in Magnesium Metal Battery Anodes

Janna Eaves-Rathert,¹ Kathleen Moyer,^{1,2} Murtaza Zohair,^{1,2} and Cary L. Pint^{1,2,3,4,*}

SUMMARY

Until now, a key barrier toward realizing a high energy density Mg^{2+} battery has been the limited understanding of mechanisms governing multivalent metal electrodeposition. This, compounded with recent observations of Mg dendrites, highlights the need for better fundamental insight into multivalent systems. We present a comprehensive study of electrodeposition in practical coin-cell configurations to evaluate the mechanisms of growth from common $\text{Mg}(\text{TFSI})_2$ -based electrolytes. Our findings indicate a transition from charge-transfer-limited to diffusion-limited electrodeposition processes that govern the morphological evolution of Mg deposits. We observe the signature of cell shorting under a wide range of current densities that we attribute to 3D hemispherical growth of Mg deposits that form under mixed diffusion and kinetic control and are distinguished from traditional fractal dendrites. Our results highlight synergy with classical electrochemical theories for growth and lay groundwork for future approaches to achieve stable electroplated multivalent metal electrodes.

INTRODUCTION

The defect that nearly destroyed the lithium (Li) battery following its debut—the dendrite—has been the target of persistent engineering efforts over the last 40 years. In the area of Li batteries, the transition to anodes incorporating metallic Li has required researchers to identify, understand, prevent, eliminate, and most recently, predict dendrites in batteries, and this remains an ongoing area of rich scientific and applied interest.¹ Now, as researchers begin to venture from studying metallic Li to alternative alkali and alkali-earth metal chemistries, the lessons learned regarding electrochemical nucleation and growth provide a blueprint from which to draw new ideas and direction. Of all alkali metals that have promise beyond Li, magnesium (Mg) stands out as it stores twice the volumetric energy as Li, and it was originally lauded for enhanced safety over Li when early studies suggested resistance to dendrite growth during electroplating.^{2–4} These properties positioned Mg rechargeable batteries (MRBs) as the ideal candidate for compact, high-performance energy storage systems relevant to electric vehicles, aerospace systems, or Internet of Things applications.

Although dendritic growth of Mg metal was reported by Gregory et al. in 1990,⁵ recent studies from the Banerjee group and the Passerini group have reminded us that Mg metal dendrites remain a threat to the future of MRBs.^{6–8} Despite strong evidence that dendrites exist, however, there remains controversy over the galvanostatic signature of failure in this and similar systems. With this said, common

Context & Scale

In an increasingly electrified world, there is a growing need for energy storage beyond lithium-ion batteries to enable future advances in sensing, computing, communications, or the Internet of Things, among others. Rechargeable multivalent magnesium (Mg) metal batteries provide an intriguing way to address this due to a higher energy density and proposed safer operation arising from smooth and dendrite-free electroplating. However, recent reports have called into question the acclaimed safety of Mg metal batteries based on observations of dendritic growth during electroplating. Moving forward, the design and development of future multivalent Mg metal batteries requires clarity regarding the mechanisms at play during electroplating over a wide range of practical operating conditions and how these impact dendrite formation or shorting events. This understanding can guide efforts to exploit their tremendous energy storage capability while mitigating processes that compromise safety.

electrolytes for Mg electroplating, such as the family of halogen-free electrolytes comprising magnesium bis(trifluoromethane sulfonyl)imide [Mg(TFSI)₂] in ether-based solvents, have been observed to exhibit galvanostatic transients featuring a high initial overpotential period, which suddenly disappears after some duration.^{6,9–12} Studies have shown that while Mg(TFSI)₂ is a popular electrolyte that is easily accessible and possesses excellent anodic stability (>3 V versus Mg/Mg²⁺), it is also highly sensitive to contamination by water and prone to Mg surface passivation.^{10,13–16} In turn, early studies attributed the high overpotential to breakdown of such surface passivating films, likely under the assumption that Mg was resistant to dendrite formation during electroplating.^{2,3} However, following recent observations of dendrite formation during Mg metal electroplating, there is a need to reevaluate growth mechanisms and understand the factors controlling non-equilibrium growth in halogen-free electrolytes.

To explain the mechanism for Mg dendrite formation, it has been shown that with high current density and poor surface properties, the electrochemical reaction rate can surpass the self-diffusion rate leading to dendritic growth of Mg.^{7,17} In addition, concentration gradients can be critical in dictating morphology, especially under constant current conditions so commonly used for the study of novel electrolytes and storage materials. Drawing a parallel to Li metal plating studies, it has been shown that electrolyte diffusion limitations are responsible for triggering the transition from root-growing mossy Li to tip-growing dendritic Li, which penetrates separators to cause catastrophic failure.^{18–22} In this area, recent work by the Dasgupta and co-workers used a combination of advanced galvanostatic techniques and microscopy to distinguish kinetic-limited growth in early cycling of a Li-Li symmetric cell from mass-transport-limited growth induced by accumulation of dead Li- in later cycles.^{22,23} The transition from kinetic- to mass-transfer-limited growth is characterized by Sand's time, τ_{sand} , which proceeds as the inverse square of current density. For times greater than the Sand's time, onset and growth of ramified deposits are well described by the models of Fleury, Chazalviel, Rosso, and colleagues, which expand upon the theory that dendrites result from diffusion limitations that create a space charge region.^{24–26} Further, depletion of cations near the interface results in preferential deposition at regions of high curvature.^{18,27–29} It is also possible to observe Sand-like behavior at unexpectedly low current densities due to inhomogeneities on the electrode surface causing local fluctuations in ion concentration.^{29–31} Three-dimensional (3D) growth due to cation depletion can in turn be avoided by increasing the electrolyte concentration,^{32,33} pulse charging,^{34,35} or homogenizing ion flux.^{36–38}

Translating such ideas to Mg is challenging since existing studies of metallic Mg electro-deposition either do not investigate the mechanistic underpinnings of shorting or study processes in harsh Grignard reagents, which are incompatible with conventional battery components and purposely push the system to extreme limits.^{7,8} In this article, we explore the regimes of the galvanostatic Mg metal electroplating curve and correlate voltage fluctuations to nucleation, growth, and shorting phenomena in practical coin-cell configurations. Notably, our findings indicate that Mg metal nucleation and deposition from TFSI-based electrolytes do occur within the high polarization region previously attributed to Mg passivation layer breakdown, leading to eventual shorting of the cell. Our work shows that intermittent resting can modify the formation and growth of 3D Mg structures, supporting the hypothesis that slow Mg diffusion leads to partial electrolyte depletion at the substrate interface. Further, although there exist more liberal definitions of the term, we exercise the strict definition of "dendrite" as a fractal structure with well-defined branches in order to distinguish these from

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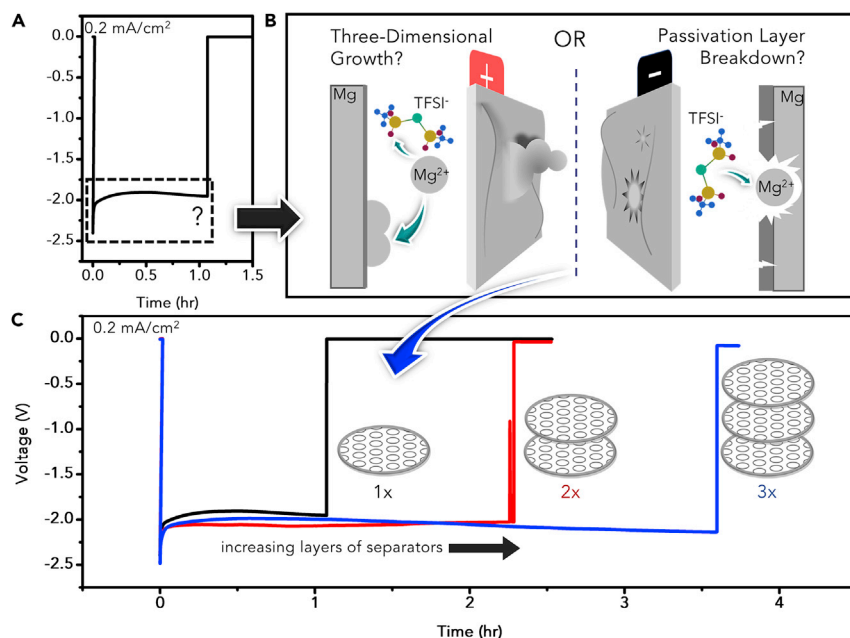


Figure 1. Electrodeposition Behavior in Mg Symmetric Cells

(A) Typical galvanostatic plating curve for a Mg-Mg symmetrical cell at 0.2 mA/cm^2 in the presence of 0.1 M Mg(TFSI)_2 in 1:1 glyme:diglyme by volume. The boxed section encloses the overpotential plateau, which is thought to be a region of either passivation layer breakdown or nucleation and growth of 3D Mg deposits.

(B) Schematic describing aforementioned processes possibly occurring during plateau region.

(C) Resulting galvanostatic profiles after varying the number of Celgard 2325 separators used in same Mg-Mg symmetrical cell configuration.

hemispherical deposits in describing Mg morphologies deposited from halogen-free electrolytes, as the growth kinetics and prevention mechanisms are contradictory to those expected of dendrites.

RESULTS AND DISCUSSION

Electrodeposition Behavior in Mg Symmetric Cells

In recent years, multiple reports have demonstrated the electroplating of Mg metal onto surfaces using galvanostatic methods with very similar electrochemical responses.^{6,9–12} The first step in our study was therefore to evaluate the electroplating behavior of Mg metal batteries under constant current discharge testing. The shape of a typical galvanostatic electroplating curve for a Mg-Mg symmetric cell is depicted in Figure 1A. Notable features include an initial spike followed by a period of negative potential, which abruptly jumps to almost 0 V after some duration. Although the precise origin is unclear, the magnitude of the high overpotential plateau has been shown to be sensitive to a number of factors, including anode and cathode surface characteristics,^{13,14} electrolyte concentration,^{39,40} or contamination due to water.^{13,41} Its sudden disappearance has been previously attributed to the first instance of Mg stripping from an inhomogeneously passivated anode^{9,11} or from the sudden removal of adsorbed species at the Mg-electrolyte interface, which are overcome in the presence of large bias.^{10,12,14} However, recent observations of Mg dendrites^{6,7} invite a third possible interpretation of this behavior; it is therefore hypothesized that stripping and plating begin with the onset of applied current and proceed throughout the plateau until 3D growth reaches the counter electrode, creating a short circuit (Figure 1B).

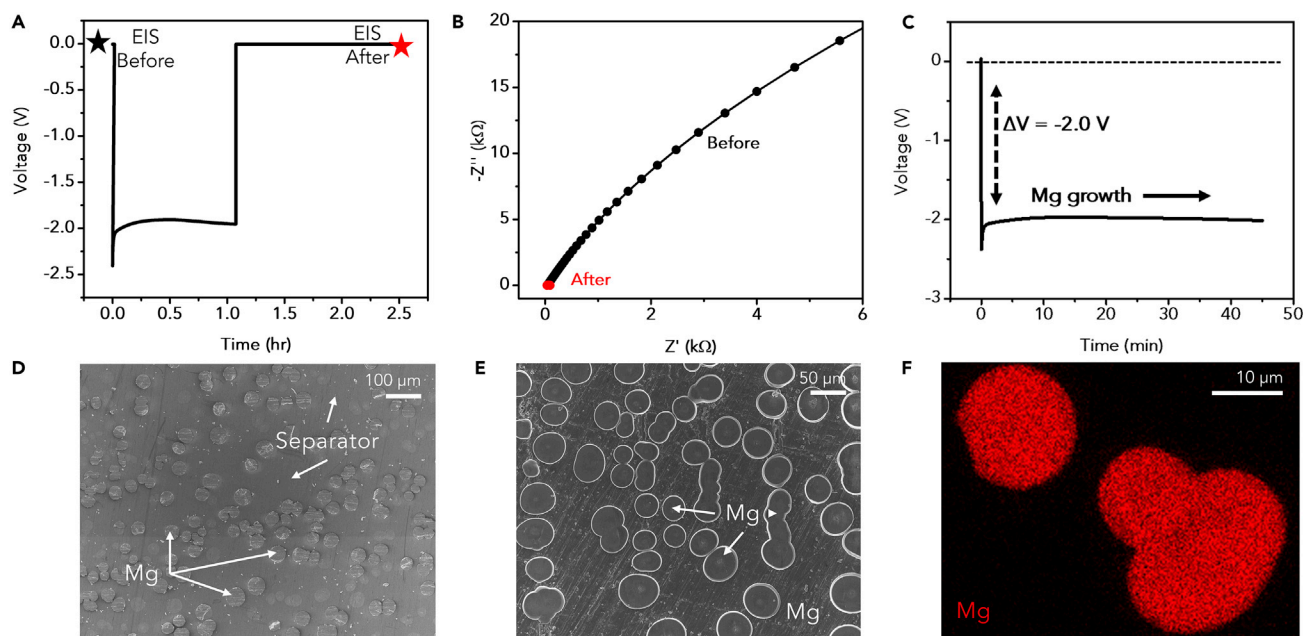


Figure 2. Identification of Short-Circuit Signature in Mg Electrodeposition

(A) Galvanostatic plating curve for a Mg-Mg symmetrical cell at 0.2 mA/cm².

(B) EIS of symmetrical cell before and after plating through high overpotential regime. See also Figure S2. Spectra acquired with 10 mV amplitude in frequency range 0.5 Hz–100 kHz.

(C) Constant current electrodeposition of Mg on Mg substrate at 0.2 mA/cm², stopped after 45 min for disassembly and SEM analysis.

(D) Electron micrograph of trilayer PP/PE/PP separator from cell subjected to constant current routine shown in (C) and peeled away from substrate, showing strong adhesion of Mg hemispheres to separator.

(E) Electron micrograph of bases of Mg deposits left behind on Mg substrate after electrochemical metal plating as in (C).

(F) EDS map of Mg, confirming that Mg is the primary element in observed deposits.

To better understand and isolate the true meaning of this plateau, we performed galvanostatic plating experiments in Mg-Mg symmetric cells with increasing layers of separators. Here, a negative current was applied such that reduction of Mg²⁺ occurs at the working electrode, resulting in a negative plating voltage. Galvanostatic plating curves from Mg-Mg cells incorporating one, two, and three separators are shown in Figure 1C. We emphasize that the same electrode and electrolyte preparation practices were employed for all three cells. Overall, the three curves shown in Figure 1C have similar features except for the duration of the plateau. Adding a second separator increased the time to breakdown by 113%, and adding a third increased time to breakdown by an additional 122%. Since the anode and electrolyte were held constant, the changing separator thickness must exert control over breakdown time. The hypothesis that the plateau represents the breakdown of a passivation or adsorption layer therefore cannot explain the strong correlation.

To better understand this observation with increasing separator thickness, each coin cell in the separator study was subjected to electrochemical impedance spectroscopy (EIS) before and after electroplating through the high overpotential regime, as shown in Figure 2A. The results for the single separator case in Figure 2B show a significant reduction in interface resistance (R_{int}) and the imaginary component of impedance. Additionally, a Warburg impedance component is expected at low frequencies for systems that undergo mass transfer, usually manifesting as a linear tail at low frequencies. Its absence implies a path for electron transfer that bypasses polarization of the electrolyte. In contrast to a perfect resistor, the system maintains

some capacitive elements, signifying an intact passivation layer at either the counter electrode surface, deposit surface, or both, qualifying the phenomenon as a soft short circuit.^{6,21,42,43} The 2- and 3-separator EIS results can be found in Figure S2.

To further elucidate the processes occurring during Mg metal plating on polished Mg substrates, a cell was stopped in the midst of the plateau region, decrimped, washed, and analyzed to identify the presence or absence of Mg deposits (Figure 2C). As shown in Figure 2D, scanning electron microscope (SEM) imaging of the separator revealed numerous polycrystalline islands extending up into the porous scaffold with undersides sheared off of the bases. The bases remained fixed to the substrate as sections of hemispheres, occasionally coalescing into dumbbells (Figure 2E) with flat topside appearance attributed to mechanical shearing during separator removal. In the specific case of electron dispersive spectroscopy (EDS), gold was used as the current collector in order to enhance contrast, but polished Mg is used as a substrate in all other experiments. Whereas gold itself may undergo limited alloying with Mg,⁴⁴ the relatively high applied current density (0.2 mA/cm²) and low temperature (298 K) prohibits diffusion into the bulk material. Supporting this hypothesis, the voltage trace possesses the same general shape as with Mg substrates, lacking any plateaus early in the plating period that might indicate a phase transformation (Figure S3). As a caveat, it is important to note that subtle differences in surface roughness, crystal structure, and nucleation sites can produce differences in deposit morphology. EDS mapping of the hemispherical deposits on gold confirmed that Mg was the primary constituent (Figure 2F), accompanied by trace fluorine and sulfur (Figure S4). The detection of Mg demonstrates conclusively that nucleation takes place “prior to” the sudden reduction in polarization at the end of the plateau. Accordingly, we conclude that after a period of growth, 3D Mg deposits protrude through the separator and produce a soft-shortening event, to which we thereby attribute the rapid decrease in overpotential in the galvanostatic discharge profile. The presence of fluorine and sulfur points to the existence of an interfacial layer on the deposits, the precise composition of which has been noted to contain MgO, Mg(OH)₂, MgCO₃, and MgF₂ in other reports.^{6,13,14} Detailed characterization of surface chemistry and the influence of operating conditions remain underexplored in the literature and require further attention from the community.

Diffusion Limitations in Mg(TFSI)₂ Electrolytes

In the study of electrodeposition, hemispherical island-type growth often results from mixed kinetic and diffusion controls.^{28,45–47} To gauge the extent of diffusion control, we measured the diffusion coefficient using an instantaneous potential step method.²⁷ From the slope of the resulting Anson plot in Figure S5, the diffusion constant, D_O , was determined to be 1.03×10^{-7} cm²/s for the 0.1 M electrolyte solution, a full order of magnitude lower than typical Li-ion electrolytes.¹⁸ It follows that with sluggish diffusion kinetics, rapid depletion can occur at the substrate-electrolyte interface during bulk electrolysis. For any current density exceeding the limiting current density, it is well understood that the interfacial cation concentration is rapidly depleted and species are reduced as quickly as they reach the surface, typically resulting in non-equilibrium morphologies, which can be dendritic in nature.^{8,18,27,29} The limiting current density, J_{lim} is defined as

$$J_{lim} = \frac{FD}{1 - t_+} \frac{2c^*}{L} \quad (\text{Equation 1})$$

where F is Faraday's constant, t_+ is the cation transference number (extrapolated to be 0.272 for a dilute solution of Mg(TFSI)₂ in diglyme³⁹), D is the diffusion coefficient,

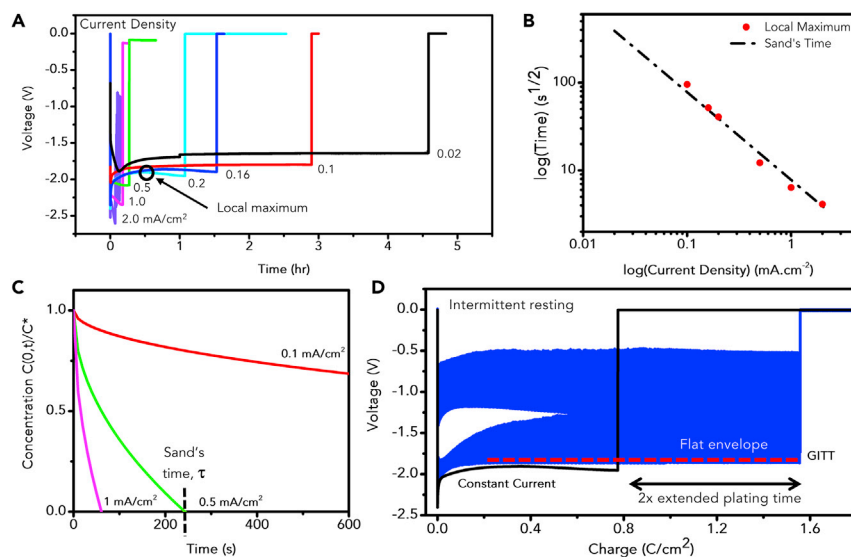


Figure 3. Diffusion Limitations in 0.1 M Mg(TFSI)₂ Electrolytes

(A) Galvanostatic transients of Mg-Mg symmetrical cells plated at current densities from 0.02 to 2 mA/cm².
(B) Local maximum of overpotential plateaus extracted from curves in (A) plotted alongside Sand's time calculated according to Equation 2.
(C) Computed concentration profiles at the substrate interface after constant current plating at 1, 0.5, and 0.1 mA/cm² for extended time periods.
(D) GITT applied to Mg-Mg symmetrical cell with current density 0.2 mA/cm² applied for 15 with 180 s of rest between pulses.

c^* is the bulk cation concentration, and L is the diffusion length. For $D = 1.03 \times 10^{-7}$ cm²/s and an approximate cation transference number of 0.272³⁹ across a 25 μ m separator, the limiting current density is about 1 mA/cm², which is within an order of magnitude of the currents typical for electrochemical analysis of Mg anode and cathode materials.

To better evaluate the transition to diffusion-limited growth at current densities within a few orders of magnitude of J_{lim} , we focus on a transition time, τ , at which the concentration of reactants at the surface, c_0 , equals zero during extended electrolysis. This can be calculated using the Sand equation, first derived by H.J.S. Sand in 1901^{27,28} and represented by

$$\tau^{\frac{1}{2}} = \frac{nFAD_O^{\frac{1}{2}}\pi^{\frac{1}{2}}c^*}{2i(1-t_+)} \quad (\text{Equation 2})$$

For times greater than τ , non-equilibrium plating is probable and other species—like solvent molecules—may be consumed by the excess electrical current. Here, n is the number of electrons transferred (assumed to be 2 for Mg²⁺), A is the electrode area, i is the applied current, and the other variables are defined as in Equation 1. Using Equation 2, we calculate Sand's time for experiments performed at 0.2 mA/cm² to be 1,519 s, or about 25 min, which corresponds well to a local voltage maximum in the high (negative) overpotential regime from data in Figure 3A, denoted with a circle. This signifies the point where decreasing concentration of reacting species at the growth front dictates a more negative reduction potential, consistent with the Nernst equation.^{27,28} To more broadly evaluate this concept, we analyzed this transition time at all current densities shown in Figure 3A ranging from 0.02 to 2.0 mA/cm² and found that the time corresponding to the minimum overpotential

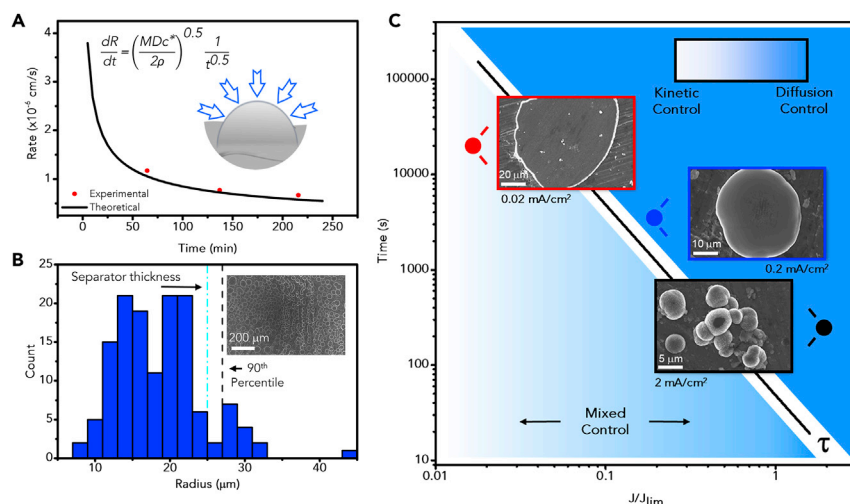


Figure 4. 3D Growth of Mg Is Moderated by Current Density

(A) Experimentally determined rate dR/dt of Mg hemispherical island growth from 0.1 M $\text{Mg}(\text{TFSI})_2$ compared with rate predicted by diffusion-controlled model of 3D growth.^{28,46}
 (B) Histogram of radii of individual Mg islands deposited on Mg substrate at 0.2 mA/cm^2 , derived from electron micrograph shown in inset. Scale bar, 200 μm .
 (C) Dependence of transition time on fraction of the limiting current density and morphologies deposited at 0.02, 0.2, and 2 mA/cm^2 as a result of competition between kinetic and diffusion control.

correlates closely with the $i^{-1/2}$ trend of transition time calculated according to Equation 2, achieving an R^2 value of 0.990 (Figure 3B). To further define what the transition time means for this system, we used the diffusion coefficient to simulate the theoretical concentration profile at the substrate surface over time for three different current densities (Figure 3C). For all the sampled current densities between 0.1 and 2 mA/cm^2 , both above and below the limiting current density, there is an experimentally apparent transition to purely diffusion-controlled growth modes preceding an ultimate shorting event. Experimentally, the relationship falls off at current densities much smaller than J_{lim} , when solid-state diffusion and reaction kinetics dominate over mass transport in the electrolyte.^{18,28,29} Besides its outlying breakdown time, the galvanostatic transient for 0.02 mA/cm^2 in Figure 3A also has a unique feature at short timescales: the voltage decreases gradually over 10 min to a minimum point (Figure S6). In Li electrodeposition, analogous behavior has been attributed to simultaneous nucleation and formation of solid electrolyte interphase, therein requiring extra charge to supply sufficient activation energy.⁴⁸ Although $\text{Mg}(\text{TFSI})_2$ -based electrolytes are not known to form solid electrolyte interphase (SEI), it is speculated that the solvent reacts with surface defects to form a passivating film instead, which could further impede self-diffusion.^{13,14,40,49}

Building from experiments so far, the influence of mass transfer limitations on the shape of the galvanostatic profile was further analyzed using galvanostatic intermittent titration technique (GITT). GITT is often used to study kinetic behavior because periodically resting the cell establishes near-static equilibrium conditions via minimization of concentration gradients through the electrolyte during short “on” periods.²² If mass transport effects play a significant role in electrochemical behavior, the envelope formed by the minimum point in each pulse should have a different shape than the steady state galvanostatic transient. GITT results are plotted in Figure 4A with respect to charge transferred, allowing for direct comparison to the

original plating curve. Using a pulse time of 15 s (much less than the transition time of 25.3 min) at 0.2 mA/cm² followed by a 3-min rest period, the GITT envelope forms an extremely flat plateau, lacking any statistically significant local maximum, which was previously established as an indication of transition to pure diffusion control. Additionally, the overpotential was slightly reduced, and the total charge passed was improved by a factor of 2.

It is evident from experiments discussed herein that Equation 2 provides insight into the mass-transfer-limited growth of Mg metal nuclei at practical current densities commonly present in electroplating studies. In this way, one can utilize Equation 2 as a blueprint to control parameters in the cell, which can postpone the reaction processes occurring beyond the transition window to mitigate 3D growth and eventual cell shorting, an idea supported by GITT measurements. Since it is generally impractical to introduce rest or re-equilibration times during battery operation, other modifications that can bear similar outcomes include increasing the transference number and modifying the electrolyte concentration, providing multiple ways to control and moderate shorting effects. Using longer chain glymes⁶ and inclusion of additives^{12,50,51} have also been effective in preventing shorting, although care must be taken not to shrink the electrochemical stability window. Further, the Coulombic efficiency of Mg stripping and plating may be improved by avoiding diffusion-limited regimes where cation depletion could lead to surface passivation via reduction of solvent, anions, or contaminants.

3D Island Growth of Mg

Despite the fact that ramified dendritic growth is not expected when a system is driven below the critical current density, the observation of soft-shortening phenomena under mild conditions emphasizes a need to unravel the true growth mechanisms dictating Mg growth. Even at current densities that are two orders of magnitude below J_{lim} , we observe an eventual soft short during electroplating. One would naturally correlate a shorting effect to be correlated to dendrite formation since this has been observed with other alkali metals, such as Li and Na, and dendrites have most recently been demonstrated to occur for Mg. In the context of metal electroplating, a dendrite is defined as a fractal structure in which the growth front proceeds at the rate of anion migration in order to preserve electroneutrality.^{18,24–26,30,52} The velocity of dendrite growth is related to the applied bias, V , by the mobility, μ ,^{27,30} which is independent of time. In contrast, revisiting the observations of Figure 1C leads to the conclusion that velocity of propagation through one, two, and three separators at 0.2 mA/cm² ($J < J_{lim}$) does not analytically match the rate of traditional dendrite propagation. In fact, as shown herein, this is more accurately modeled as the growth of a 3D island under diffusion control (Figure 4A). Taking the growth velocity as the total separator thickness divided by the time spent under diffusion control, we find that the velocity decreases with time according to the classic solution for 3D hemispherical growth under diffusion control^{28,46}:

$$\frac{dR}{dt} = \left(\frac{MDc^*}{2\rho} \right)^{0.5} \frac{1}{t^{0.5}} \quad (\text{Equation 3})$$

where R is the radius for a hemisphere with 90° wetting angle, M is the molar mass of Mg, ρ is the density of Mg, and the other variables are as previously defined in Equations 1 and 2. This result is surprising because it implies that the electrodeposited metallic Mg was unperturbed by the presence of multiple trilayer polypropylene/polyethylene/polypropylene (PP/PE/PP) separators and in fact used the separator as a scaffold for growth. This scaffolding effect is immediately clear from SEM imaging studies of Mg deposited at 0.2 mA/cm² through glass fiber separators

(Figure S7), which preferentially break apart before shearing the deposit itself. In cross section, the islands are pictured engulfing the fibers, while a top view from the perspective of the counter electrode shows them penetrating all the way through. Some possible reasons for wetting of the separator by zerovalent Mg could be high Mg surface tension,¹⁷ or the complex rearrangement of solvation structures throughout the process of multiple electron transfer.^{49,53} This implies that the 3D propagation of Mg metal growth could at least in part be moderated by separators with controlled surface properties.⁵⁴

To further analyze the Mg morphology that forms under different current densities and understand these observations, we imaged deposits formed on Mg substrates at 0.02, 0.2, and 2 mA/cm². Figure 4B shows a histogram of 137 individual islands grown at 0.2 mA/cm², demonstrating that 90% of deposits have a radius less than 25 μm , which is the thickness of one separator. Assuming a general hemispherical model where the height is equal to the radius, this would mean that the cell short circuited when some of these deposits grew to $R = 25 \mu\text{m}$, cutting off the driving force for any further reduction. We attribute this hemispherical growth mechanism of Mg deposits to diffusion control, and we delineate this from dendrite formation that would otherwise yield a similar shorted device outcome. At 0.02 mA/cm²—much lower than the critical current density—the bases of deposits left behind after separator removal deviate from circular geometry and grew significantly larger before shorting, indicating enhanced kinetic control. At 2 mA/cm²—above the limiting current density—we find a multitude of small nuclei that pile on top of each other like caterpillar segments, realizing a more traditional dendrite conformation since significant diffusion limitation drives preferential nucleation at protrusions. The average nucleus size and density follow trends consistent with those established for galvanostatic deposition of Li metal, where higher current densities resulted in smaller, more numerous spherical nuclei that uniformly cover the substrate.⁴⁸ Direct comparison is presented in Figure S8. Overall, SEM images showing representative deposits formed at these three current densities are included as insets in Figure 4C, demonstrating the competition between charge transfer and mass transfer at current densities within 2 orders of magnitude of the critical current density, J_{lim} . At very low current densities applied in the Mg(TFSI)₂-ether system, strong interactions between adjacent Mg adatoms¹⁷ over large nuclei lead to 3D growth, which eventually traverses the separator to cause a soft short at times less than the transition time, τ , calculated according to Equation 2. At moderate current densities, mixed control gives way to mass transport limitation at times greater than τ , and deposits behave according to diffusion-controlled 3D island growth. For high current densities $J > J_{\text{lim}}$, the concentration of cations is depleted at the surface almost instantaneously and protruding deposits of stacked spheres rapidly bridge the separator. In all cases, electrolysis concludes with a soft-shortening event. By applying SEM analysis alongside GITT and traditional electrochemical models of nucleation and growth, we highlight a comprehensive picture of 3D growth that brings together past observations drawing from dendrite formation occurring due to solid-state and/or surface diffusion mechanisms as well as our results that highlight 3D growth controlled by sluggish transport in commonly used electrolytes.

Conclusion and Outlook

Overall, our results paint a picture that demonstrates key advances in understanding of Mg metal electroplating. First, our results indicate that the origin of the high overpotential commonly observed in Mg metal electroplating occurs subsequent to Mg metal nucleation, meaning that this cannot be attributed to common arguments of passivation layer removal. Second, our results highlight a competition between

different mechanisms for electroplating over a full range of current densities practical to Mg battery operation—none of which form smooth layer-by-layer deposits that were previously assumed to be the case for electrodeposited Mg layers. Third, our results distinguish uniform 3D growth of Mg scaffolded by the separator from that of traditional, ramified electrodeposits known as “dendrites” —both of which rely on different mechanisms but lead to the same symptomatic result in a battery of a soft-shortening effect. Although extended electrolysis tests have been extremely rare in Mg electrolyte studies to date, we suspect that 3D growth is not limited to TFSI-based electrolytes; anomalous voltage changes in early cycling accompanied by major reduction in impedance and/or SEM images of hemispherical growth have also been reported for TFSI with triphenylborohydride additive,⁵⁰ Mg(CB₁₁H₁₂)₂ (MMC),^{10,55} Mg(HMDS)₂,⁵¹ and Mg[B(HFIP)₄]₂ electrolytes.⁵⁶

Moving forward, understanding the different mechanisms behind Mg growth enables modification of the process. In the regime of 3D hemispherical growth at moderate current densities, one can leverage the insight of solution transport properties to control the mechanism of growth to some degree. Further, in the regime dictated by solid state phenomena and surface diffusion,^{3,7,17} one can seed the surface with abundant nuclei using a short high-overpotential pulse, or moderate surface properties of the current collector in a manner that has been recently explored for Na and Li metal batteries.^{57–59} Finally, the weakness of traditional separators in impeding Mg growth may catalyze conversation about unique engineering solutions for multivalent systems and possibly open the door to solid electrolytes as a practical option for Mg,⁶⁰ despite pervasive transport challenges in solid-state systems. Collectively, we assert that understanding and manipulating the strong electrostatic interactions characteristic of multivalent ions, and the mechanisms distinguishing observed behavior from that of monovalent alkali metals like Li and Na, are key steps toward realizing next-generation energy dense storage media.

EXPERIMENTAL METHODS

Resource Availability

Lead Contact

Further information and requests for resources and materials should be directed to and will be fulfilled by the Lead Contact, Cary L. Pint (carypint@iastate.edu).

Materials Availability

This study did not generate new unique materials.

Data and Code Availability

The datasets supporting the current study are available from the Lead Contact on request.

Electrochemical Testing

Galvanostatic testing was conducted in SS316 coin cells using an MTI BTS8-MA 8-channel battery analyzer using 0.1 M Mg(TFSI)₂ electrolyte in 1:1 v/v diglyme/dimethoxyethane. Mg(TFSI)₂ salt was purchased from Solvionic and dried under vacuum (1E-2 torr) overnight at 250°C. Diglyme and dimethoxyethane (DME) (Sigma Aldrich) were dried over 30% m/V molecular sieves for 48 h before electrolyte preparation in an argon glovebox (< 1 ppm O₂). Electrolytes were used within 48 h of preparation. Electrolyte quality was assessed via cyclic voltammetry using polished Mg foil counter and reference electrodes, a polished Pt disc working electrode with diameter 0.25 mm, and a scan rate of 20 mV/s from –1 to 3V. The first cycle Coulombic efficiency was determined to be 32.6% with plating and stripping

peak onsets of -0.65 and 1.68 V, respectively (Figure S1). This result is consistent with other studies that measured water content less than 20 ppm.^{39,41} For coin cells and three-electrode tee cells, 0.1-mm-thick Mg foil (MTI) was punched into 8 mm discs and polished using 220, 400, and 1,000 grit sandpaper sequentially. Celgard 2325 trilayer separators were dried under vacuum overnight before inclusion in coin cells or tee cells. EIS was performed on coin cells prepared in the same manner using a Metrohm Autolab PG810 with FRA32M frequency response analyzer in the frequency range 0.1 MHz–0.5 Hz with amplitude 10 mV.

Morphological Analysis

Coin cells were de-crimped, the electrode or separator of interest rinsed twice in dry DME, and stored in a sealed vial for transport to SEM. Samples were exposed to air during transfer. Imaging was conducted using a Zeiss Merlin SEM with EDS detector.

Measurement of Diffusion Coefficient

A potential step experiment was performed in a sealed, flooded, 3-electrode cell using polished Mg foil counter and reference electrodes, a polished Pt disc working electrode with diameter 0.25 mm. The voltage was set to -0.5 V versus Mg/Mg²⁺ for 300 s, then the resulting current was integrated over time to produce a line with slope related to the diffusion coefficient as defined by the Cottrell equation:^{27,40}

$$i(t) = \frac{nFAD_O^{\frac{1}{2}}C_O^*}{\pi^{\frac{1}{2}}} \left(\frac{1}{t^{\frac{1}{2}}} \right) \quad (\text{Equation 4})$$

The transference number is extrapolated from data acquired by Sa et al., using atomic absorption spectroscopy for t_+ versus concentration of Mg(TFSI)₂ in diglyme with a standard deviation of 0.108. The impact of potential error in transference number on fit to experimental results is depicted graphically in Figure S9.

SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at <https://doi.org/10.1016/j.joule.2020.05.007>.

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AUTHOR CONTRIBUTIONS

Conceptualization, J.E.-R. and C.L.P.; Methodology, J.E.-R., K.M., M.Z., and C.L.P.; Investigation, J.E.-R.; Writing – Original Draft, J.E.-R. and C.L.P.; Writing – Review & Editing, J.E.-R. and C.L.P.; Funding Acquisition, J.E.-R. and C.L.P.; Resources, C.L.P.; Supervision, C.L.P.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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